

Transformers and Attention-Based Deep Networks

Assignment - 1

Graduate School of Informatics

Due: 22.03.2026

Introduction

The objective of this assignment is to increase familiarity with attention-based (transformer) architectures and their applications. **The task is to classify the sentiment of customer service conversations.** This is a two-week assignment that should not be rushed in the final days. Any submission after the due date will have a penalty of 10 points for each day. You will be unable to submit after 2 days.

Input: Customer-support interaction. **Output:** 3 Classes for **positive**, **negative** or **neutral**

Sentiment Analysis

1. Set up Weights and Biases (WANDB), create an account if you don't have one yet. WANDB is an experiment tracking tool that can also be used to automated reporting and hyperparameter tuning.
2. Set up GitHub account for version control. Version control is a mandatory skill for any software developer.
3. Download the dataset. Note that there are train.csv and test.csv. Do not leak any data from test.csv, it is a very small portion of the original dataset. The dataset is a customer service interaction data where customers define their problems and agents of the company try to help them.
4. Prepare an Exploratory Data Analysis for the dataset, indicating the distribution of the sentiment classes and other important facts about the data. Is there any correlation between other features and the sentiment feature? This helps with understanding the data and provides useful insight into the problem.
5. Determine necessary features and prepare your train and validation subsets. Do not leak data from the test set.
6. Data pre-processing: Explain which pre-processing steps you have taken for the dataset and justify your approaches.
7. Train (or fine-tune) a suitable attention-based network for sentiment classification.
8. Evaluate the trained model on the test set.
9. Report the results using appropriate evaluation metrics for this task.

Deliverables

Submit to ODTUCLASS in a single zip file with the naming convention "assignment_1.STUDENTNUMBER.zip".

Report: Prepare a report that clearly explains the steps in this assignment.

Code: The code that is to be shared must be working and reproducible, submit your code to odtuclass.

Version Control: Make your code publicly available at your [github](#) repo.

Experiment Tracking: Record experiments in this assignment in your public [WANDB](#) dashboard or report.

Format

The report document must be a single pdf file. The report shall be in IEEE format. You can find the IEEE template [here](#) (Conferences, Original Research, Word or Latex). Software must be shared as an interactive notebook (.ipynb) with outputs on! If you are using any online interactive notebook service (e.g. Kaggle or Google-Colab) for your coding part, you can download it as a notebook with your code and markdown text.

Dataset

The dataset is available [here](#). There are two files for the dataset, train.csv for training and validation, test.csv to be used only for testing and not to be leaked into the training!

Report

The report shall contain the following:

Abstract: Prepare an abstract (100 words max)

Introduction: Brief introduction about the submission.

Dataset: Explain and present pre-processing steps and exploratory data analysis briefly.

Modeling: Explain the model of your choice, the modifications you made and their benefits for the task.

Evaluation: Discuss which evaluation metrics were used in the assignment and why did you choose them.

Results: Share the results of the experiments as a summary of WANDB experiment tracking tool.

Discussion: Discuss your findings and results.

Conclusion: Conclusion and final remarks.

Grading

1. **Data exploration and pre-processing (20 pts):** You shall report the pre-processing steps you have taken in this assignment. You shall also prepare a concise and brief exploratory data analysis, figures, tables and plots to augment your report.
2. **Version control (Git) (10 pts):** your GitHub repository shall be public and contain meaningful commits. Do not dump your code to Git, your repository should show your incremental development.
3. **Experiment Tracking (WANDB) (10 pts):** your wandb experiments or report page shall be public. If it is not public you will not get any points from experiment tracking.
4. **Model training (40 pts):** Follow clean coding principles. If the code is messy and hard to understand you will not get full points.
5. **Report (20 pts):** Your report shall be **2 page maximum**, if you exceed this limitation you will get -10 points for each extra page. Your report is your only way of communication for your assignment, it shall explain clearly all necessary steps you have taken.